

Lecture 15

Fundamentals of Numerical Analysis

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Next Topics

- 1 Interpolation: Approximating a function $f(x)$ by a polynomial $p_n(x)$.
- 2 Differentiation: Approximating the derivative of a function $f(x)$.
- 3 Integration: Approximating an integral $\int_a^b f(x) dx$

Interpolation: Introduction

Objective

Approximate an unknown function $f(x)$ by an easier function $g(x)$, such as a polynomial.

Objective (alt)

Approximate some data by a function $g(x)$.

Types of approximating functions:

- 1 Polynomials
- 2 Piecewise polynomials
- 3 Rational functions
- 4 Trig functions
- 5 Others (inverse, exponential, Bessel, etc)

Interpolation: Introduction

How do we approximate $f(x)$ by $g(x)$? In what sense is the approximation a good one?

- 1 Interpolation: $g(x)$ must have the same values of $f(x)$ at set of given points.
- 2 Least-squares: $g(x)$ must deviate as little as possible from $f(x)$ in the sense of a 2-norm: minimize $\int_a^b |f(t) - g(t)|^2 dt$
- 3 Chebyshev: $g(x)$ must deviate as little as possible from $f(x)$ in the sense of the ∞ -norm: minimize $\max_{t \in [a,b]} |f(t) - g(t)|$.

Interpolation: Introduction

Given $n + 1$ distinct points $x_0, \dots, x_n$, and values $y_0, \dots, y_n$, find a polynomial $p(x)$ of degree n so that

$$p(x_i) = y_i \quad i = 0, \dots, n$$

- A polynomial of degree n has $n + 1$ degrees-of-freedom:

$$p(x) = a_0 + a_1x + \dots + a_nx^n$$

- $n + 1$ constraints determine the polynomial uniquely:

$$p(x_i) = y_i, \quad i = 0, \dots, n$$

Theorem

If points $x_0, \dots, x_n$ are distinct, then for arbitrary $y_0, \dots, y_n$, there is a *unique* polynomial $p(x)$ of degree at most n such that $p(x_i) = y_i$ for $i = 0, \dots, n$.

How can you prove the polynomial is unique? (Hint: What if it isn't?)

Monomials

Obvious attempt: try picking

$$p(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$$

So for each x_i we have

$$p(x_i) = a_0 + a_1x_i + a_2x_i^2 + \cdots + a_nx_i^n = y_i$$

OR

$$a_0 + a_1x_0 + a_2x_0^2 + \cdots + a_nx_0^n = y_0$$

$$a_0 + a_1x_1 + a_2x_1^2 + \cdots + a_nx_1^n = y_1$$

$$a_0 + a_1x_2 + a_2x_2^2 + \cdots + a_nx_2^n = y_2$$

$$a_0 + a_1x_3 + a_2x_3^2 + \cdots + a_nx_3^n = y_3$$

$$\vdots$$

$$a_0 + a_1x_n + a_2x_n^2 + \cdots + a_nx_n^n = y_n$$

Monomial: The problem

$$\begin{bmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ 1 & x_2 & x_2^2 & \dots & x_2^n \\ & & & \vdots & \\ 1 & x_n & x_n^2 & \dots & x_n^n \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

Question

- Is this a “good” system to solve?

Example

from Recktenwald

Consider Gas prices (in cents) for the following years:

x	year	1986	1988	1990	1992	1994	1996
y	price	133.5	132.2	138.7	141.5	137.6	144.2

```
1 year =[1986    1988    1990    1992    1994    1996 ]';
2 price=[133.5    132.2    138.7    141.5    137.6    144.2]';
3
4 M = vander(year);
5 a = M\price;
6
7 x=linspace(1986,1996,200);
8 p=polyval(a,x);
9 plot(year,price,'o',x,p,'-');
```

Back to the basics...

Example

Find the interpolating polynomial of least degree that interpolates

x	1.4	1.25
y	3.7	3.9

Directly

$$\begin{aligned}p_1(x) &= \left(\frac{x - 1.25}{1.4 - 1.25} \right) 3.7 + \left(\frac{x - 1.4}{1.25 - 1.4} \right) 3.9 \\&= 3.7 + \left(\frac{3.9 - 3.7}{1.25 - 1.4} \right) (x - 1.4) \\&= 3.7 - \frac{4}{3}(x - 1.4)\end{aligned}$$

Lagrange

What have we done? We've written $p(x)$ as

$$p(x) = \left(\frac{x - x_1}{x_0 - x_1} \right) y_0 + \left(\frac{x - x_0}{x_1 - x_0} \right) y_1$$

- the sum of two linear polynomials
- the first is zero at x_1 and 1 at x_0
- the second is zero at x_0 and 1 at x_1
- these are the two linear Lagrange basis functions:

$$\ell_0(x) = \frac{x - x_1}{x_0 - x_1} \quad \ell_1(x) = \frac{x - x_0}{x_1 - x_0}$$

Lagrange

Example

Write the Lagrange basis functions for

x	$\frac{1}{3}$	$\frac{1}{4}$	1
y	2	-1	7

Directly

$$\ell_0(x) = \frac{(x - \frac{1}{4})(x - 1)}{(\frac{1}{3} - \frac{1}{4})(\frac{1}{3} - 1)}$$

$$\ell_1(x) = \frac{(x - \frac{1}{3})(x - 1)}{(\frac{1}{4} - \frac{1}{3})(\frac{1}{4} - 1)}$$

$$\ell_2(x) = \frac{(x - \frac{1}{3})(x - \frac{1}{4})}{(1 - \frac{1}{3})(1 - \frac{1}{4})}$$

Lagrange

The general Lagrange form is

$$\ell_k(x) = \prod_{i=0, i \neq k}^n \frac{x - x_i}{x_k - x_i}$$

The resulting interpolating polynomial is

$$p(x) = \sum_{k=0}^n \ell_k(x) y_k$$

Example

Find the equation of the parabola passing through the points (1,6), (-1,0), and (2,12)

$$x_0 = 1, x_1 = -1, x_2 = 2; \quad y_0 = 6, y_1 = 0, y_2 = 12;$$

$$\begin{aligned}\ell_0(x) &= \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} = \frac{(x+1)(x-2)}{(2)(-1)} \\ \ell_1(x) &= \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} = \frac{(x-1)(x-2)}{(-2)(-3)} \\ \ell_2(x) &= \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} = \frac{(x-1)(x+1)}{(1)(3)}\end{aligned}$$

$$\begin{aligned}p_2(x) &= y_0\ell_0(x) + y_1\ell_1(x) + y_2\ell_2(x) \\ &= -3 \times (x+1)(x-2) + 0 \times \frac{1}{6}(x-1)(x-2) \\ &\quad + 4 \times (x-1)(x+1) \\ &= (x+1)[4(x-1) - 3(x-2)] \\ &= (x+1)(x+2)\end{aligned}$$

Summary so far:

- Monomials: $p(x) = a_0 + a_1x + \cdots + a_nx^n$ results in poor conditioning
- Monomials: but evaluating the Monomial interpolant is cheap (nested iteration)
- Lagrange: $p(x) = \ell_0(x)y_0 + \cdots + \ell_n(x)y_n$ is very well behaved.
- Lagrange: but evaluating the Lagrange interpolant is expensive (each basis function is of the same order and the interpolant is not easily reduced to nested form)

Fixing Monomials, Fixing Lagrange

Back to the gas price example. Suppose we use a better basis like

$$(x - \bar{x})^k$$

instead of

$$x^k$$

For example, $\bar{x} = \text{average}(x_i), i = 0, \dots, n$.

The basis $(x - \bar{x})^k$ are called *shifted monomials* because x is shifted by $\bar{x}$.

Recall Nested Form

Given a polynomial

$$p(x) = -5 + 4x - 7x^2 + 2x^3 + 3x^4$$

we can write this as

$$p(x) = -5 + x(4 + x(-7 + x(2 + 3x)));$$

evaluation can be done from the inside-out, for cheap (nested evaluation).
This polynomial can also be written as

$$p(x) = -5 + 2x - 4x(x - 1) + 8x(x - 1)(x + 1) + 3x(x - 1)(x + 1)(x - 2)$$

in nested form

$$p(x) = -5 + x(2 + (x - 1)(-4 + (x + 1)(8 + 3(x - 2))))$$

Newton Polynomials

- Newton Polynomials are of the form

$$p_n(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1) + a_3(x - x_0)(x - x_1)(x - x_2) + \dots$$

- The basis used is thus

function	order
1	0
$x - x_0$	1
$(x - x_0)(x - x_1)$	2
$(x - x_0)(x - x_1)(x - x_2)$	3

- More stable than monomials
- More computationally efficient (nested iteration) than using Lagrange and shifted monomials

Newton Polynomials using Divided Differences

Consider the data

x_0	x_1	x_2
y_0	y_1	y_2

We want to find a_0 , a_1 , and a_2 in the following polynomial so that it fits the data:

$$p_2(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1)$$

Matching the data gives three equations to determine our three unknowns a_i :

at x_0 : $y_0 = a_0 + 0 + 0$

at x_1 : $y_1 = a_0 + a_1(x_1 - x_0) + 0$

at x_2 : $y_2 = a_0 + a_1(x_2 - x_0) + a_2(x_2 - x_0)(x_2 - x_1)$

Newton Polynomials using Divided Differences

Or in matrix form:

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & x_1 - x_0 & 0 \\ 1 & x_2 - x_0 & (x_2 - x_0)(x_2 - x_1) \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} y_0 \\ y_1 \\ y_2 \end{bmatrix}$$

⇒ lower triangular

⇒ only $\mathcal{O}(n^2)$ operations

Question

How many operations are needed to find the coefficients in the monomial basis?

Newton Polynomials using Divided Differences

Using Forward Substitution to solve this lower triangular system yields:

$$\begin{aligned}a_0 &= y_0 = f(x_0) \\a_1 &= \frac{y_1 - a_0}{x_1 - x_0} \\&= \frac{f(x_1) - f(x_0)}{x_1 - x_0} \\a_2 &= \frac{y_2 - a_0 - (x_2 - x_0)a_1}{(x_2 - x_1)(x_2 - x_0)} \\&= \frac{f(x_2) - f(x_0) - (x_2 - x_0)\frac{f(x_1) - f(x_0)}{x_1 - x_0}}{(x_2 - x_1)(x_2 - x_0)} \\&= \frac{\frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0}\end{aligned}$$

Newton Polynomials using Divided Differences

From this we see a pattern. There are many terms of the form

$$\frac{f(x_j) - f(x_i)}{x_j - x_i}$$

These are called *divided differences* and are denoted with square brackets:

$$f[x_i, x_j] = \frac{f(x_j) - f(x_i)}{x_j - x_i}$$

Applying this to our results:

$$a_0 = f[x_0]$$

$$a_1 = f[x_0, x_1]$$

$$\begin{aligned} a_2 &= \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0} \\ &= f[x_0, x_1, x_2] \end{aligned}$$

Newton Polynomials using Divided Differences

example: long way

Example

For the data

x	1	-4	0
y	3	13	-23

Find the 2nd order interpolatin polynomial using Newton.

We know

$$p_1(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1)$$

And that

$$a_0 = f[x_0] = f[1] = f(1) = 3$$

$$a_1 = f[x_0, x_1] = \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{13 - 3}{-4 - 1} = -2$$

$$a_2 = f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0}$$

$$= \frac{\frac{-23-13}{0-4} - \frac{13-3}{-4-1}}{0-1}$$

$$= \frac{-9+2}{-1} = 7$$

So

$$p_1(x) = 3 - 2(x - 1) + 7(x - 1)(x + 4)$$

Divided Differences

Recursive Property

$$f[x_0, \dots, x_k] = \frac{f[x_1, \dots, x_k] - f[x_0, \dots, x_{k-1}]}{x_k - x_0}$$

With the first two defined by

$$\begin{aligned} f[x_i] &= f(x_i) \\ f[x_i, x_j] &= \frac{f[x_j] - f[x_i]}{x_j - x_i} \end{aligned}$$

Divided Differences

Invariance Theorem

$f[x_0, \dots, x_k]$ is invariant under all permutations of the arguments $x_0, \dots, x_k$

Simple “proof”: $f[x_0, x_1, \dots, x_k]$ is the coefficient of the x^k term in the polynomial interpolating f at $x_0, \dots, x_k$. But any permutation of the x_i still gives the same polynomial.

This says that we can also write

$$f[x_i, \dots, x_j] = \frac{f[x_{i+1}, \dots, x_j] - f[x_i, \dots, x_{j-1}]}{x_j - x_i}$$

Divided Differences

the easy way: tables

We can compute the divided differences much easier using tables. To construct the divided difference table for $f(x)$ for the $x_0, \dots, x_3$

x	$f[\cdot]$	$f[\cdot, \cdot]$	$f[\cdot, \cdot, \cdot]$	$f[\cdot, \cdot, \cdot, \cdot]$
x_0	$f[x_0]$			
x_1	$f[x_1]$	$f[x_0, x_1]$	$f[x_0, x_1, x_2]$	
x_2	$f[x_2]$	$f[x_1, x_2]$	$f[x_1, x_2, x_3]$	$f[x_0, x_1, x_2, x_3]$
x_3	$f[x_3]$	$f[x_2, x_3]$		

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x_0	$f[x_0]$			
x_1	$f[x_1]$	$f[x_0, x_1]$	$f[x_0, x_1, x_2]$	$f[x_0, x_1, x_2, x_3]$
x_2	$f[x_2]$	$f[x_1, x_2]$	$f[x_1, x_2, x_3]$	
x_3	$f[x_3]$	$f[x_2, x_3]$		

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x_0	$f[x_0]$			
x_1	$f[x_1]$	$f[x_0, x_1]$		
x_2	$f[x_2]$	$f[x_1, x_2]$	$f[x_0, x_1, x_2]$	
x_3	$f[x_3]$	$f[x_2, x_3]$	$f[x_1, x_2, x_3]$	$f[x_0, x_1, x_2, x_3]$

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		$f[x_0, x_1]$		
x_1	$f[x_1]$		$f[x_0, x_1, x_2]$	
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		$f[x_0, x_1]$		
x_1	$f[x_1]$		$f[x_0, x_1, x_2]$	
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x_2	$f[x_2]$	$f[x_1, x_2]$	$f[x_1, x_2, x_3]$	$f[x_0, x_1, x_2, x_3]$
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		$f[x_2, x_3]$		
x_3	$f[x_3]$			

Divided Differences

the easy way: example

Construct the divided differences table for the data

$$\begin{array}{ccccc} x & 1 & \frac{3}{2} & 0 & 2 \\ y & 3 & \frac{13}{4} & 3 & \frac{5}{3} \end{array}$$

and construct the largest order interpolating polynomial.

We can compute the divided differences much more easily using tables. To construct the divided difference table for $f(x)$ for the $x_0, \dots, x_3$

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$\frac{3}{2}$	$\frac{13}{4}$	$\frac{1}{2}$	$\frac{1}{3}$	
0	3	$\frac{1}{6}$	$-\frac{5}{3}$	-2
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$\frac{3}{2}$	$\frac{13}{4}$	$\frac{1}{2}$	$\frac{1}{3}$	
0	3	$\frac{1}{6}$	$-\frac{5}{3}$	-2
2	$\frac{5}{3}$	$-\frac{2}{3}$		

The coefficients are readily available and we arrive at

$$p_2(x) = 3 + \frac{1}{2}(x-1) + \frac{1}{3}(x-1)\left(x - \frac{3}{2}\right) - 2(x-1)\left(x - \frac{3}{2}\right)x$$